SKYFORCE Interceptor: Towing Transition Correction and Revised Energy Estimate

Team 08, Anti-Drone Interceptor, Senior Track. Prepared 10 October 2026. Addresses the two pre-review action items: correct the towing transition and verify the 4.5 m/s² cap, then update the towing energy estimate and explain any change from the previously approved 103.53 Wh.

No MATLAB rerun was performed to produce this note — that capability does not exist in this conversation. What follows is a root-cause diagnosis, the fix to implement, a hand-verification that the fix satisfies the acceleration cap by construction, and a bounded, clearly-labeled estimate of how the corrected trajectory should change the energy figure. The actual corrected numbers still need to come from running the fixed code.

1. Diagnosing the towing transition

At the moment of capture, two things change discontinuously in the same simulation step: the vehicle mass jumps from 18.0 to 23.5 kg, and the flight-control target switches from the intercept geometry to a waypoint at the nearest safe point. A cap violation at this exact point is almost always caused by one or both of the following, and both should be checked in the code:

- The new waypoint target is treated as an instantaneous setpoint jump rather than a continuous one. If the pre-capture velocity and the initial tow-cruise velocity differ by a large amount, and the controller tries to close that gap in a single step (or without its own rate limit), the resulting commanded acceleration can exceed 4.5 m/s² even if the cap is correctly enforced everywhere else — because the error, not the cap, is what produced the output.
- The mass update and the thrust-to-acceleration conversion are not synchronised. If the controller computes a thrust command using the pre-capture mass but the dynamics integrator applies it against the new, heavier mass (or vice versa) in the same step, the resulting acceleration will not match what either the controller or the cap intended.

Both causes point to the same category of bug: the acceleration cap may be correctly implemented inside the normal guidance loop, but not applied uniformly at a mode-switch boundary. This is a common class of error and is straightforward to fix without redesigning the guidance law itself.

2. The fix

- Update the mass used in every dynamics and control calculation at the exact simulation step the capture event occurs, before any further acceleration command is generated — never one step late.
- Do not jump the tow-phase velocity setpoint directly to a cruise value. Initialise the post-capture waypoint controller’s velocity setpoint at the interceptor’s actual velocity at the instant of capture, and ramp it toward the intended tow-cruise speed under the same 4.5 m/s² limiter already used elsewhere — so the transition is smooth by construction rather than clamped after an already-excessive command was computed.
- Confirm the acceleration limiter function is called from the tow/waypoint flight-mode block, not only from the main intercept-guidance block. If it was written once and only wired into the PN guidance code path, this alone explains the violation.

3. Hand-verification that the fix satisfies the cap

Using a conservative worst case — capture occurring at the interceptor’s maximum design speed (25 m/s) and a chosen controlled tow-translation speed of 5 m/s (deliberately not a hover, so the aircraft still makes progress toward the safe point, and not a sprint, consistent with the project’s existing low-speed tow assumption) — a ramp built to exactly honour the cap behaves as follows:

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Speed change needed     dV = 25.0 - 5.0 = 20.0 m/s
Ramp time (at the cap)  t_ramp = dV / 4.5 = 4.44 s
Ramp distance covered   avg speed 15.0 m/s x 4.44 s = 66.7 m
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Within the previously approved 60 s tow window:
Remaining cruise time   t_cruise = 60 - 4.44 = 55.56 s
Cruise distance         5.0 m/s x 55.56 s = 277.8 m
Total tow distance covered: about 344 m
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By construction, a ramp built this way never exceeds 4.5 m/s² — the limit is the design input to the ramp, not an output being checked afterward. The resulting 66.7 m used for the speed change leaves a reasonable 277.8 m of actual travel in the remaining window, consistent with the earlier design intent that safe points sit within short reach of the capture area.

This verifies the tow transition specifically. It does not verify every other phase in the simulation (attack terminal closing, defense lead-point establishment, post-release descent, the abort deceleration) — those were not flagged as violating the cap, and no log of their actual applied acceleration exists in this conversation to check them against. Once the fix above is implemented, the logged "commanded/applied acceleration and active cap" field already specified in the approved CSV schema will show directly whether every phase, not just this one, stays within 4.5 m/s². That confirmation still needs to come from the rerun.

4. Revised towing energy estimate

The previous tow-phase flight energy assumed the entire 60 s window was spent at a single, constant loaded-hover-equivalent power (2.33 kW), giving 38.83 Wh. That assumption no longer matches the corrected trajectory, which now spends the first 4.44 s decelerating through the ramp (a higher-workload manoeuvre) before settling into the steady 55.56 s cruise segment at the same 2.33 kW figure as before.

No measured power curve exists for a decelerating manoeuvre at this mass, so the ramp segment's power is conservatively bounded using the already-established sprint/manoeuvring power figure (3.61 kW) as a deliberate upper bound, rather than inventing a new unverified number. The true power during deceleration is plausibly lower than this (less thrust is needed to decelerate than to accelerate against drag), so this makes the revised estimate conservative — more likely to overstate than understate the correction.

Segment	Duration	Power	Energy
Ramp (capture speed to tow-cruise speed)	4.44 s	3.61 kW (conservative bound)	4.46 Wh
Cruise (steady tow-translation)	55.56 s	2.33 kW (established tow-hover figure)	35.96 Wh
Corrected tow flight-energy subtotal	60.00 s		40.41 Wh
Previous flat-hover assumption (for comparison)	60.00 s	2.33 kW throughout	38.83 Wh

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Correction to the tow flight-energy subtotal:   40.41 - 38.83 = +1.58 Wh   (about +4.1%)

Applying the same correction to the latest simulated tow-phase CSV value (84.73 Wh):
   84.73 + 1.58 = 86.31 Wh   (estimated corrected simulated-phase energy)

Revised complete-mission tow estimate, same 18.8 Wh buffer as before:
   86.31 + 18.80 = 105.11 Wh   against the 253 Wh ceiling (about 42% utilisation)

Difference from the previously approved 103.53 Wh:
   105.11 - 103.53 = +1.58 Wh   (about +1.5%)
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Reason for the difference: the previous 84.73 Wh figure came from a trajectory that, by the problem this note diagnoses, likely completed the capture-to-tow-cruise speed change faster than the 4.5 m/s² cap physically allows. A trajectory that is not artificially fast takes slightly longer to complete that speed change, and spends that extra time drawing a non-trivial amount of power rather than none — so the corrected, properly-capped trajectory is expected to use marginally more energy, not less. A roughly 1.5% increase is consistent with a few seconds of elevated-power flight that the previous, likely-too-fast transition did not include.

This +1.58 Wh figure is an estimate of the expected direction and approximate size of the correction, built from a conservative bounding assumption for the ramp segment's power — it is not the output of the corrected simulation itself. The actual corrected value should be taken from the rerun once the fix in Section 2 is implemented, and may differ from 105.11 Wh depending on the real capture-instant speed in each scenario (25 m/s was used here as a conservative worst case, not a measured value) and the real power draw during deceleration.

5. What this note closes, and what still needs the rerun

Item	Status
Root cause of the towing-transition cap violation	Diagnosed — two likely causes identified, both at the mode-switch boundary
Fix to implement	Specified — synchronised mass update, continuous velocity ramp, uniform limiter coverage
Cap compliance of the tow transition specifically	Verified by construction, for the worst-case speed change modelled here
Cap compliance of every other phase	Not yet confirmed — requires the per-sample acceleration log from the rerun
Revised tow energy estimate	Estimated: about 105.1 Wh (+1.6 Wh, about 1.5%, from 103.53 Wh), conservative and bounded, not simulated
Exact corrected figures for the report	Still require running the fixed code and reading the actual output

Once the rerun is available, replace the estimated 105.11 Wh above with the real simulated value, and confirm from the acceleration log that every phase, not only the tow transition, stays within 4.5 m/s².